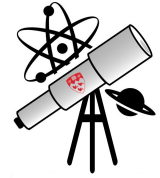




McGill Physical Journal



Type the Title of your Paper, Capitalize First Letter

First Author¹, Second Author², and Third Author¹

¹First affiliation, Address, City and Postcode, country

²Second affiliation, Address, City and Postcode, Country

Abstract

Maximum 250 words. Times New Roman 12 pt. font.

In scientific journals, the Abstract is meant to tell a potential reader whether or not the article contains information worth reading in full. Therefore, the Abstract presents a comprehensive factual summary using a minimum number of words. This summary should include a short (i.e. 1 - 2 sentences!) description of:

1. what was done,
2. how it was done (experimental technique used with the relevant experimental conditions),
3. what were the final results (including the error estimate),
4. what are the main conclusions (e.g., if the results are consistent with a particular model, literature data, etc.).

It should be kept in mind that since the Abstract is read together with the title of the report any information given in the title need not be repeated in the Abstract.

1. Introduction

Maximum 500 words. Times New Roman 12 pt. font.

This section outlines basic goal of the experiments and the relevant theory used for interpreting/analyzing the results should appear first in the main body of the report. The purpose of the Introduction is to present the background information, basic concepts and equations required by the reader to understand i) the purpose of the experiment and ii) the methods used in treating the data.

It serves little purpose, and is not considered good practice, to rewrite or copy large portions from notes, the lab manual or texts. Furthermore, derivations of equations given elsewhere should not be reproduced. If the derivation is of particular interest reference can be made to the appropriate source or it can be placed in an appendix. However, a brief presentation of basic theory, with relevant equations, in the words and style of the author is essential.

Your work will likely build on, rely on, or otherwise make use of the results or work of others. In all such cases it is essential that you properly reference the appropriate original source. **References should be numbered consecutively in order of first appearance throughout the text.** Software is available to the McGill community to make this very easy (EndNote for Window/Word, BibTeX for LaTeX) which is available to all McGill students. References to Web URL's are to be discouraged but are sometimes necessary. This is an example of a reference to a peer-reviewed journal article [1], a book [2], online material [3], and a technical document such as a Laboratory manual [4]. For papers with more than five authors include only the first author's name followed by 'et al.'.

See additional information on this and other sections in the lecture notes.

2. Methods

Maximum 500 words. Maximum 2 Figures. Times New Roman 12 pt. font.

A description of the equipment and procedures used to make the measurements are given in this section. When special apparatus is used, a description of the operating principles together with a clearly labelled schematic diagram is desirable.

Experimental procedure should include a concise outline of the actual procedure, which should differ somewhat from that described in the notes or elsewhere. This section should NOT be a detailed transcription of the manual or take the form of a "recipe"; e.g. "and then we pressed 'setup 2' on the oscilloscope, and then moved the cursor to measure the position of...". Instead, it should be a very concise outline of the main points of the procedure sufficient to convey a clear overview of the experimental procedure to the reader. This outline should consist of concise factual statements, e.g., "The heat capacity of the calorimeter and contents were determined by measuring the temperature change that resulted from the addition of a known quantity of heat".

3. Results

Maximum 500 words. Use the minimum number of Figures and Tables necessary to fully present your results. Times New Roman 12 pt. font.

This section presents the experimental data and the results of any analyses including the estimation of experimental uncertainties. **Your results should be presented in a logical sequence.**

Do not jump around in an inconsistent, hard to follow manner. Any assumptions made in the treatment of data should be clearly specified. The reader should be informed, using explanatory sentences how the raw data obtained in the laboratory were transformed into the desired results; i.e. your graphs and tables should appear in this section, but this section is not ONLY your graphs and tables.

4. Discussion

Maximum 500 words. Use the minimum number of Figures and Tables necessary to fully discuss your results. Times New Roman 12 pt. font.

The Discussion evaluates whether or not the experiment objectives have been met (i.e. was the experiment a success) and what the results actually mean. This portion of the report allows some freedom, but the purpose is to explain the significance/meaning of the results to the reader; i.e. to “interpret” the results.

In the discussion section, important features of the results should be highlighted. Comparison should be made with accepted values if relevant and available. This comparison gives an estimate of the accuracy of the experiment and provides an opportunity to assess different methods (i.e. those you used and those others have used) and to comment on your error analysis. **Do the results obtained in this experiment agree - within quoted uncertainty - of accepted values? If there is a discrepancy, by how many standard deviations? Why?**

What (if any) is the theoretical significance of the results? The Discussion should include a clear, concise presentation of how the results demonstrate key concepts or impact on questions of theoretical importance.

Any direct questions asked of you in the lab manual should also be addressed in this section.

5. Conclusions

Maximum 300 words. Times New Roman 12 pt. font.

In the Conclusion section, state the most important outcome of your work. Do not simply summarize the points already made in the body, instead, interpret your findings at a higher level of abstraction. Show whether, or to what extent, you have succeeded in addressing the need stated in the Introduction. At the same time, do not focus on yourself (for example, by restating everything you did). Rather, show what your findings mean to readers. Make the Conclusion interesting and memorable for them.

At the end of your Conclusion, consider including perspectives, that is, an idea of what could or should still be done in relation to the issue addressed in the paper. If you include perspectives,

clarify whether you are referring to firm plans for yourself and your colleagues (“In the coming months, we will . . .”) or to an invitation to readers (“One remaining question is . . .”).

References

- [1] S. Tan, M. Pustilnik, and L. I. Glazman, “Relaxation of a high-energy quasiparticle in a one-dimensional bose gas,” *Phys. Rev. Lett.* *105*, no. 105, 2010. 2
- [2] I. Hughes and T. Hase, *Measurements and Their Uncertainties: A Practical Guide to Modern Error Analysis*. Oxford, UK: Oxford Univ. Press, 2010. 2
- [3] J. Manaster, “Sloth squeak,” <http://blogs.scientificamerican.com/psi-vid/2014/04/09/sloth-squeak>, accessed: 2022-10-31. 2
- [4] B. Ruffolo and B. Vachon, “Phys-257 lab-x manual,” McGill University, Tech. Rep., 2022. 2

Author Contributions Statement

All manuscripts must include an Author Contributions Statement detailing the contribution of each author. Author names must be listed as initials. For example, “A.B and C.D wrote the main manuscript text and E.F prepared figures 1-3. All authors reviewed the manuscript.”

Appendix A. An example appendix

Authors including an appendix section should do so after References section.

The relevant sample calculations for the data tables and figures should appear in an appendix

A.1. Example of a sub-heading within an appendix